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advantage: 1. 因切成多個區間，運算量大幅下降， $N \log N \rightarrow N$
 2. 硬體簡單（不管多長，FFT點數固定）
 只需一種晶片 #

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$$h[n] = 0.09x[n-2] + 0.36x[n-1] + 0.55x[n] + 0.55x[n+1] + 0.36x[n+2] + 0.09x[n+3]$$

$$= 0.09 \{ x[n-2] + 4(x[n-1] + x[n+2]) + x[n+3] \} + 0.55(x[n+1] + x[n])$$

$\Rightarrow 2N$ muls if $x[n]$ is real #

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(a) direct

$$3NM = 3 \times 1500 \times 250 = 1125000 \text{ #}$$

Non-Sectioned Convolution

$$P \geq N+M-1 = 1749, \text{ 取 } 2016$$

$$2M \times 2016 + 3 \times 2016 = 31504 \text{ #}$$

Section Convolution

$$L_0 = 2106, D_0 = 2106 + 250 - 1 = 2355$$

$$P = 2304: L = 2304 - 250 + 1 = 2055, S = \left\lceil \frac{1500}{2055} \right\rceil = 1$$

$$S \times (2M \times 2304 + 3 \times 2304) = 38648$$

$$P = 2520: L = 2520 - 250 + 1 = 2271, S = \left\lceil \frac{1500}{2271} \right\rceil = 1$$

$$S \times (2M \times 2520 + 3 \times 2520) = 40640$$

$$P = 2048: L = 2048 - 250 + 1 = 1799, S = \left\lceil \frac{1500}{1799} \right\rceil = 1$$

$$S \times (2M \times 2048 + 3 \times 2048) = 39816$$

$$P = 2016: L = 2016 - 250 + 1 = 1767, S = \left\lceil \frac{1500}{1767} \right\rceil = 1, S \times (2M \times 2016 + 3 \times 2016) = 31504 \text{ #}$$

Ans:

(i) Non-Sectioned Convolution or section Conv.

(ii) $P = 2016$

(iii) 31504 #

(b) direct

$$3NM = 3 \times 1500 \times 50 = 225000 \#$$

Sectioned Conv.

$$L_0 = 324, P_0 = 324 + 50 - 1 = 373.$$

P	L	S	Mul.
312	263	6	24912
336	287	6	22992
360	311	5	20800
420	371	5	27100

→ #

Non-Sectioned Conv.

$$P \geq 1549, \text{ 取 } 1680$$

$$2 \text{ Mul}_{1680} + 3 \times 1680 = 52100 \#$$

Ans: (i) Sectioned Conv.

(ii) $p = 360$

(iii) $20800 \#$

(c) direct

$$3NM = 3 \times 1500 \times 10 = 45000 \#$$

Sectioned Conv.

$$L_0 = 42, P_0 = 42 + 10 - 1 = 51$$

P	L	S	Mul
45	31	42	19950
48	39	39	12792
52	43	35	20020
54	45	34	21012
56	47	32	15360

→ #

Non-Sectioned Conv

$$P \geq 1509, \text{ 取 } 1680$$

$$2 \text{ Mul}_{1680} + 3 \times 1680 = 52100 \#$$

Ans:

(i) Sectioned Conv.

(ii) $p = 48$

(iii) $12792 \#$

(d) direct

$$3NM = 3 \times 1500 \times 2 = 9000$$

Sectioned Conv.

$$L_0 = 2, P_0 = 2 + 2 - 1 = 3.$$

P	L	S	Mul
2	1	1500	9000
3	2	150	9750
4	3	500	6000
5	4	375	13125

→ #

Non-Sectioned Conv

$$P \geq 1504, \text{ 取 } 1680$$

$$2 \text{ Mul}_{1680} + 3 \times 1680 = 52100 \#$$

Ans:

(i) Sectioned Conv.

(ii) $p = 4$

(iii) $6000 \#$

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(Walsh 之變化較劇烈)

(a) 不行, Walsh 只能用在 logic convolution, 不能用在 linear convolution. #

(b) 可以, Walsh transform 可用在 step function, Walsh 之變化適用 #
 4 性 變 易

6

(a) $32 \times \log_2 32 = 160$ #

(b) $\frac{16}{2} + 14 = 30$ Adds #
 16 point add $\rightarrow$ 8 points

7

(a) $[1, 0, 1] \rightarrow [1, +1]$, $[1, 1, 0] \rightarrow [1, -1]$, $[0, 1, 1] \rightarrow [-1, 1]$

$W_1 = [1111, 1111, 1111, 1111]$, $W_4 = [1, +1, -1, -1, 1, -1, -1, 1]$

$W_0 = [1, -1, 1, -1, 1, -1, 1, -1]$

$[1, -1] \rightarrow [1111, 1111, 1111, 1111]$
 $+1+1, +1+1, +1+1, +1+1$
 $111, 1111, 1111, 1111]$

$[1, -1] \rightarrow [1, -1, 1, -1, 1, -1, 1, -1]$
 $1, -1, 1, -1, 1, -1, 1, -1$
 $1, -1, 1, -1, 1, -1, 1, -1]$

$[-1, 1] \rightarrow [-1, 1, -1, 1, -1, 1, -1, 1]$
 $-1, 1, -1, 1, -1, 1, -1, 1$
 $-1, 1, -1, 1, -1, 1, -1, 1]$

相加: $[1, 1, -3, 1, 1, -3, 3, -1, 1, 1, 3, -1, 1]$
 $1, -3, 1, 1, -3, 1, 1, -3, 1, 1, 3, -1, 3, 1$
 $1, 1, 3, -1, 1, 1, 3, -1, 3, 1, 1, -3, 1, 1]$ #

(b) 可以, missing 5th and 15th 可以設 $C > 0$ 為 1, $C < 0$ 為 -1, 依然可以還原. #

missing: $[1, 1, 3, 0, 1, 3, 3, -1, 1, 3, -1, 0, 1, -1, -3, 1, 1, -3, 1, 1, 3, 1, 1, -3, 1, 1]$ #

$$C_{11} = \frac{\langle \hat{x}, w_1 \rangle}{\langle w, w \rangle} = \frac{14}{16} > 0 \Rightarrow 1$$

(c) 不好, 因為 basis of Haar 不是完全正, 如果 data miss 一部分, 無法在 demodulation 還原. #

Bonus (150/9) $N=32$ points 6th row of H_{32} 是什麼?

[0000 0000 1111 +-1++ 0000 0000 0000 0000]
#